The documentation and process conversion measures necessary to comply with this document shall be completed by 22 June 2019.

INCH-POUND
MIL-PRF-19500/718A
22 March 2019
SUPERSEDING
MIL-PRF-19500/718
12 December 2003

PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, DIODE, SILICON, BIPOLAR TRANSIENT VOLTAGE SUPPRESSOR, TYPES 1N6950 THROUGH 1N6986, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for 5,000 watt, peak, pulse power, silicon, transient, voltage suppressor diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device as specified in MIL-PRF-19500.

* 1.2 Physical dimensions. The device package for the encapsulated device type is as follow: in accordance see figure 1.

1.3 Maximum ratings. Maximum ratings are as shown in maximum test ratings herein (see 3.7), and as follows:

$P_{M(AV)}$ = 5.0 W (derate at 33.35 mW/°C above $T_A$ = +25°C) (see figure 2, and 6.4).

-65°C ≤ $T_j$ ≤ +175°C (ambient), -65°C ≤ $T_{STG}$ ≤ +175°C (ambient).

$P_{PP}$ = 5,000 W (see figure 3) at $t_p$ = 10/1000 µs (see figure 4).

1.4 Primary electrical characteristics at $T_A$ = +25°C. Primary electrical characteristics are as shown in maximum test ratings herein (see 3.7).

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil

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* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

* 1.5.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

* 1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

* 1.5.2.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "6950" through "6986".

* 1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

## 2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contact.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

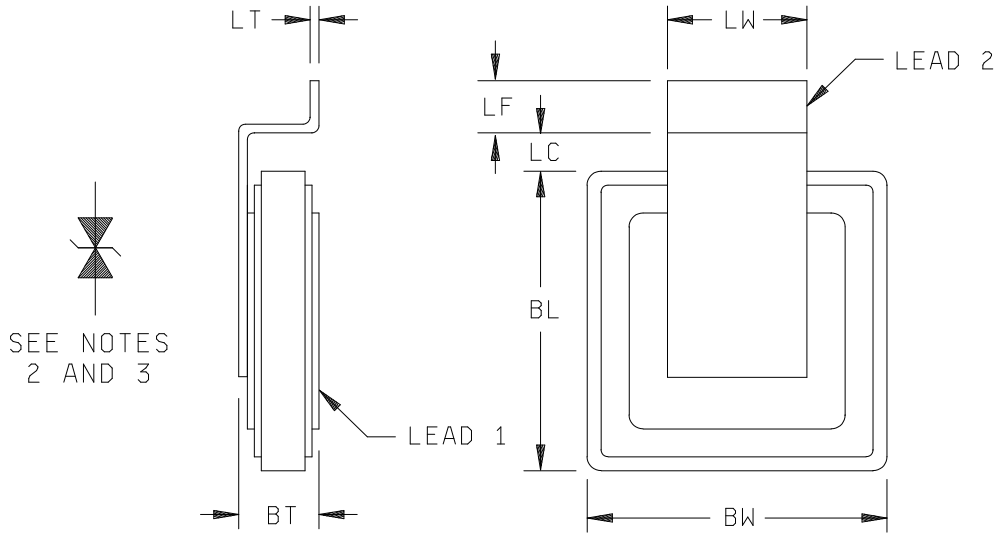
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BW | .420 | .440 | 10.67 | 11.0 | 1 |
| BL | .420 | .440 | 10.67 | 11.0 | 3 |
| BT | .085 | .105 | | | |
| LW | .195 | .205 | 4.95 | 5.21 | |
| LT | .010 | .015 | 0.25 | 0.38 | 5 |
| LF | .070 | .080 | 1.78 | 2.03 | |
| LC | .040 | .070 | 1.02 | 1.78 | 4 |

NOTES:
1. Dimensions are in inches.
2. Symbol for internal construction of bipolar transient suppressor.
3. The bottom of the case is lead 1 and the gull wing is lead 2.
4. In accordance with ASME Y14.5M.

FIGURE 1. Physical dimensions.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Metallurgical bond construction. Metallurgically bonded construction is required. The bonding flow shall have flow points above 260°C.

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I subgroup 2, table II, and table III.

3.7 Maximum test ratings. Test ratings for the devices specified herein shall be as shown in table IV.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.1.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500, and as specified herein except lot accumulation shall be 3-months in lieu of 6-weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein shall be performed on the first inspection lot to this revision to maintain qualification.



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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 3 | $T_{(high)}$ = +175°C | $T_{(high)}$ = +175°C |
| 9, 10, 11 | Not applicable | Not applicable |
| 12 | See 4.5.1 | See 4.5.1 |
| 13 | Interim electrical, delta, and table 1, subgroups 2 and 3, electrical parameters herein not applicable for this screen (performed in screen 12). | Interim electrical, delta, and table 1, subgroups 2 and 3, electrical parameters herein not applicable for this screen (performed in screen 12). |

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herin.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1056<br>1051 | -55°C to 100°C, 25 cycles, n = 22, c = 0.<br>-55°C to +175°C, 100 cycles, n = 22, c = 0. Surge not applicable. |
| B4 | | Not applicable. |
| B5 | 1027 | Condition for accelerated steady-state operation life are as follows: See 4.5.2, $T_A$ = +100°C (min); $T_J$ = +225°C (min); t = 168 hours minimum. |
| B6 | | Not applicable. |

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.



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4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1056 | -55°C to 100°C, 10 cycles, n = 22, c = 0. |
| | 1051 | -55°C to 175°C, 25 cycles, n = 22, c = 0. Surge not applicable. |
| B3 | 1027 | See 4.5.1, 1 ms pulse only (see 4.5.2). |
| B5 | | Not applicable |

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Lead tension: Test condition A; weight = 5 pounds; t = 15 ±3 s. |
| C3 | | Not applicable |
| C5 | | Not applicable |
| C6 | 1026 | See 4.5.1, (see 4.5.2). |
| C7 | 1018 | |
| C8 | | Conditions for temperature coefficient of breakdown voltage are as follows: $I_{(BR)}$ = column 3 of table IV, $T_1$ = +25°C ±3°C, $T_2$ = $T_1$ +100°C; n = 22, c = 0. |
| C9 | | Conditions for maximum peak pulse current are as follows: See 4.5.3.a, (20 μs pulse only) 10 pulses; n = 22, c = 0. |

NOTE: All electrical measurements and tests shall be performed twice, once in each direction.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See table III for delta limits when applicable.



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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Power burn-in (HTRB) and steady-state operation life test conditions. The test conditions and order of events shall be as follows:

a. Pulse in accordance with 4.5.3.b, in polarity A 10 times (screening and group B) and 50 times (group C) at $T_A$ = +25°C.

b. Pulse in accordance with 4.5.3.b, in polarity B 10 times (screening and group B) and 50 times (group C) at $T_A$ = +25°C.

c. Read $I_D$ in polarities A and B at $T_A$ = +25°C, remove defective devices and record the number of failures.

d. Apply the working peak reverse voltage ($V_{WM}$) (column 4 of table IV) at polarity A at $T_A$ = +150°C as follows:

(1) Forty-eight hours (JANTX and JANTXV) and 120 hours (JANS) for the screening test.

(2) One hundred seventy hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

(3) Five hundred hours for group C steady-state operation life test.

e. Read $I_D$ in polarity A at $T_A$ = +25°C. Devices with $\Delta I_D$ > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater shall be considered defective. Remove defective devices and record the number of failures. 1/

f. Apply the working peak reverse voltage ($V_{WM}$) (column 4 of table IV) at polarity B at $T_A$ = +150°C as follows:

(1) Forty eight hours (JANTX and JANTXV) and 120 hours (JANS) for the screening test.

(2) One hundred seventy hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

(3) Five hundred hours for group C steady-state operation life test.

g. Read $I_D$ in polarity B at $T_A$ = +25°C. Devices with $\Delta I_D$ > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater shall be considered defective. Remove defective devices and record the number of failures. 1/

h. Read $V_{(BR)}$ in polarities A and B at $T_A$ = +25°C. Devices with $\Delta V_{(BR)}$ > 2 percent (± 5 percent for steady-state operation life) of the initial reading shall be considered defective. Remove defective devices and record the number of failures. 1/

i. Read $I_D$ in polarity A at $T_A$ = +25°C, remove defective devices and record the number of failures.

4.5.1.1 Group C steady-state operation life test (alternate procedure). When the group B, 340-hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.5.1 with the following exceptions:

a. In 4.5.1, steps a and b shall be moved and performed following step g.

b. In 4.5.1, steps e and g shall be repeated after step a and b are performed and before step i is completed (step i may be omitted when this procedure is used.



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4.5.2 Accelerated steady-state operation life (JANS). This test shall be conducted with the devices subjected to the breakdown current specified in column 10 of table IV in opposite polarities for 84 + 8, - 4 hours in each polarity. At the beginning of the test and at the end of each time period, the devices shall be temperature stabilized at $T_A$ = +25°C) and subjected to pulse conditions at the rate of one pulse per minute (max) for 10 pulses in accordance with 4.5.3 as specified.

4.5.3 Maximum peak pulse current ($I_{pp}$). The peak pulse currents specified in column 7 of table IV shall be applied while simultaneously maintaining a bias voltage of not less than the applicable voltage specified in column 4 of table IV, in the same polarity as the peak pulse current. The peak pulse current shall be applied with a current vs time waveform as follows (1 pulse per minute maximum):

a. Pulse current shall reach 100 percent of $I_{PP}$ at $t \leq 8$ μs and decay to 50 percent of $I_{PP}$ at $t \geq 20$ μs for (see figure 5).

b. Pulse current shall reach 100 percent of $I_{PP}$ at $t \leq 10$ μs and decay to 50 percent of $I_{PP}$ at $t \geq 1$ ms for $t_p$ = 1 ms (see figure 4).

4.5.4 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent. The peak clamping voltage as specified in column 6 of table IV shall be applicable to the 1 ms pulse of 4.5.3.b only.

1/ For the purpose of this test, the direction in which the device is first pulsed shall be considered polarity A and the reverse direction polarity B.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits 2/ | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 3/ | | | | | | |
| Standby current | 4016 | DC method, $V_R = V_{WM}$ (column 4 of table IV) | $I_D$ | | Column 5 | µA dc |
| Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle $\leq 2$ percent, $I_{(BR)}$ = column 3 of table IV | $V_{(BR)}$ | Column 2 | Column 2 | V dc |
| Subgroup 3 3/ | | | | | | |
| Minimum breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle $\leq 2$ percent, $I_{(BR)}$ = column 3 of table IV, $T_A = -55°C$ | $V_{(BR)}$ | Column 9 | | V dc |
| Subgroup 4 3/ | | | | | | |
| Clamping voltage maximum (pulsed) (see 4.5.4) | | $t_p = 10/1000$ µs (see 4.5.3.b), $I_{PP}$ = column 7 of table IV | $V_C$ | | Column 6 | V (pk) |
| Subgroups 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table IV.
3/ All electrical testing shall be performed twice, once in each direction.



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TABLE II. Group E inspection (all quality levels) for qualification only.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Thermal shock | 1056 | 500 cycles, condition A, 0°C to +100°C. | |
| Temperature cycling | 1051 | 500 cycles, condition C, -65°C to +175°C. | |
| Electrical measurements | | See table III, steps 1, 2, and 5. | |
| Subgroup 2 | | | 22 devices<br>c = 0 |
| Life test | | 1,000 hours, see 4.5.1. | |
| Electrical measurements | | See table III, steps 1, 2, 3, 4, and 5. | |
| Subgroup 3 | 2101 | | 3 devices<br>c = 0 |
| Destructive physical analysis | | | |
| Subgroup 4 | | | |
| Thermal impedance curves | | Each supplier shall submit their (typical) design thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | |
| Peak pulse current | | See 4.5.3. $I_{PP}$ shall be characterized by the supplier and this data shall be available to the government. Test shall be performed on highest and lowest voltage in each structurally identical grouping. Test until failure occurs on all devices within the chosen sample size to characterize each voltage, or until the rated $I_{PP}$ is exceeded by 30%. | |
| Electrical measurements | | See table III, steps 1 and 2. | |
| Subgroup 7 | | | 45 devices<br>c = 0 |
| Soldering heat | 2031 | 1 cycle | |



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TABLE III. Groups B and C electrical and delta measurements. 1/ 2/ 3/ 4/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits 5/ | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Standby current | 4016 | DC method, $V_R = V_{WM}$ column 4 of table IV | $I_D$ | | Column 5 | μA dc |
| 2. | Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table IV | $V_{(BR)}$ | Column 2 | Column 2 | V dc |
| 3. | Standby current | 4016 | DC method; $V_R = V_{WM}$ (column 4 of table IV) | $\Delta I_D$ 6/ | | 100 percent of initial reading or 20 percent of column 5 of table IV, whichever is greater. | |
| 4. | Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table IV | $\Delta V_{(BR)}$ 6/ | | ± 5 percent of initial value | |
| 5. | Clamping voltage | | $t_p$ = 10/1000 μs (see 4.5.3.b); $I_{PP}$ = column 7 of table IV | $V_C$ | | Column 6 | V (pk) |

1/ All electrical testing shall be performed twice, once in each direction.
2/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, steps 1, 2 and 5.
   b. Subgroup 5, see 4.5.1.
3/ The electrical measurements for table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2.
   b. Subgroup 3, see 4.5.1.
   c. Subgroup 6 see table III herein, steps 1, 2, 3 and 4.
4/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2 and 9, see table III herein, steps 1, and 2.
   b. Subgroup 6, see 4.5.1.
5/ Column references are to table IV.
6/ Devices which exceed the table I limits for this test shall not be accepted.



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TABLE IV. Characteristics and ratings.

| Col 1 | Col 2 | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | | Col 8 | Col 9 | Col 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Type | Breakdown voltage $V_{(BR)}$ at $I_{(BR)}$ | Test current $I_{(BR)}$ | Working peak reverse voltage $V_{WM}$ | Maximum standby current $I_D$ | Maximum clamping voltage $V_C$ at $I_{PP}$ | Maximum peak pulse current ($I_{PP}$) $t_p$ = 20 µs $t_r$ = 8 µs | Maximum peak pulse current ($I_{PP}$) $t_p$ = 1 ms $t_r$ = 10 µs | Maximum temperature coefficient of $V_{(BR)}$ $\alpha V_{(BR)}$ | Minimum breakdown voltage at $I_{(BR)}$ $T_A$ = -55°C | Breakdown current $I_{(BR)}$ Maximum dc current $T_A$ = +25°C |
| | Min V dc | mA dc | V (pk) | µA dc | V (pk) | A (pk) | A (pk) | α/°C | V dc | mA dc |
| 1N6950 | 7.13 | 10 | 6.0 | 1000 | 11.3 | 2500 | 440 | 0.061 | 6.66 | 625.0 |
| 1N6951 | 7.79 | 10 | 7.0 | 500 | 12.1 | 2333 | 413 | 0.065 | 7.24 | 575.0 |
| 1N6952 | 8.65 | 10 | 7.5 | 200 | 13.4 | 2100 | 373 | 0.068 | 8.01 | 520.0 |
| 1N6953 | 9.50 | 1 | 8.5 | 50 | 14.5 | 1950 | 343 | 0.073 | 8.75 | 475.0 |
| 1N6954 | 10.50 | 1 | 9.0 | 10 | 15.6 | 1817 | 320 | 0.075 | 9.65 | 430.0 |
| 1N6955 | 11.40 | 1 | 10.0 | 5 | 16.7 | 1700 | 300 | 0.078 | 10.40 | 365.0 |
| 1N6956 | 12.40 | 1 | 11.0 | 5 | 18.2 | 1550 | 273 | 0.081 | 11.30 | 350.0 |
| 1N6957 | 14.30 | 1 | 12.0 | 5 | 21.2 | 1333 | 237 | 0.084 | 13.00 | 315.0 |
| 1N6958 | 15.20 | 1 | 13.0 | 5 | 22.5 | 1250 | 223 | 0.086 | 13.70 | 295.0 |
| 1N6959 | 17.10 | 1 | 15.0 | 5 | 25.2 | 1117 | 198 | 0.088 | 15.40 | 265.0 |
| 1N6960 | 19.00 | 1 | 17.0 | 5 | 27.7 | 1018 | 180 | 0.090 | 17.10 | 235.0 |
| 1N6961 | 20.90 | 1 | 18.0 | 5 | 30.6 | 917 | 163 | 0.092 | 18.80 | 215.0 |
| 1N6962 | 22.80 | 1 | 20.0 | 5 | 33.2 | 850 | 150 | 0.094 | 20.50 | 195.0 |
| 1N6963 | 25.70 | 1 | 22.0 | 5 | 37.5 | 750 | 133 | 0.096 | 23.00 | 175.0 |
| 1N6964 | 28.50 | 1 | 25.0 | 5 | 41.4 | 683 | 120 | 0.097 | 25.50 | 155.0 |
| 1N6965 | 31.40 | 1 | 28.0 | 5 | 45.7 | 617 | 110 | 0.098 | 29.00 | 140.0 |
| 1N6966 | 34.20 | 1 | 30.0 | 5 | 49.9 | 567 | 100 | 0.098 | 30.50 | 130.0 |
| 1N6967 | 37.10 | 1 | 33.0 | 5 | 53.9 | 518 | 93.3 | 0.100 | 33.10 | 120.0 |
| 1N6968 | 40.90 | 1 | 36.0 | 5 | 59.3 | 483 | 84.3 | 0.101 | 36.40 | 110.0 |
| 1N6969 | 44.70 | 1 | 40.0 | 5 | 64.8 | 433 | 77.3 | 0.101 | 39.80 | 100.0 |
| 1N6970 | 48.50 | 1 | 43.0 | 5 | 70.1 | 400 | 71.3 | 0.102 | 43.10 | 90.0 |
| 1N6971 | 53.20 | 1 | 47.0 | 5 | 77.0 | 367 | 65 | 0.103 | 47.30 | 85.0 |
| 1N6972 | 58.90 | 1 | 53.0 | 5 | 85.0 | 333 | 59 | 0.104 | 52.30 | 75.0 |
| 1N6973 | 64.60 | 1 | 58.0 | 5 | 92.0 | 300 | 54.3 | 0.104 | 57.30 | 70.0 |
| 1N6974 | 71.30 | 1 | 64.0 | 5 | 103.0 | 273 | 48.7 | 0.105 | 63.20 | 60.0 |
| 1N6975 | 77.90 | 1 | 70.0 | 5 | 113.0 | 250 | 44.3 | 0.105 | 69.00 | 55.0 |
| 1N6976 | 86.50 | 1 | 75.0 | 5 | 125.0 | 227 | 40 | 0.106 | 76.50 | 50.0 |
| 1N6977 | 95.00 | 1 | 82.0 | 5 | 137.0 | 207 | 36.7 | 0.106 | 84.10 | 47.5 |
| 1N6978 | 105.00 | 1 | 94.0 | 5 | 152.0 | 183 | 33 | 0.107 | 92.80 | 42.5 |
| 1N6979 | 114.00 | 1 | 100.0 | 5 | 168.0 | 167 | 29.7 | 0.107 | 100.00 | 37.5 |
| 1N6980 | 124.00 | 1 | 110.0 | 5 | 182.0 | 157 | 27.3 | 0.107 | 109.00 | 35.0 |
| 1N6981 | 143.00 | 1 | 128.0 | 5 | 213.0 | 133 | 23.3 | 0.108 | 126.00 | 30.0 |
| 1N6982 | 162.00 | 1 | 145.0 | 5 | 245.0 | 120 | 20.3 | 0.108 | 143.00 | 27.5 |
| 1N6983 | 171.00 | 1 | 150.0 | 5 | 261.0 | 113 | 19 | 0.108 | 151.00 | 26.0 |
| 1N6984 | 181.00 | 1 | 160.0 | 5 | 278.0 | 107 | 18 | 0.108 | 157.00 | 25.0 |
| 1N6985 | 190.00 | 1 | 170.0 | 5 | 294.0 | 100 | 17 | 0.108 | 169.00 | 23.5 |
| 1N6986 | 209.00 | 1 | 185.0 | 5 | 328.0 | 83 | 15.3 | 0.108 | 184.00 | 21.5 |



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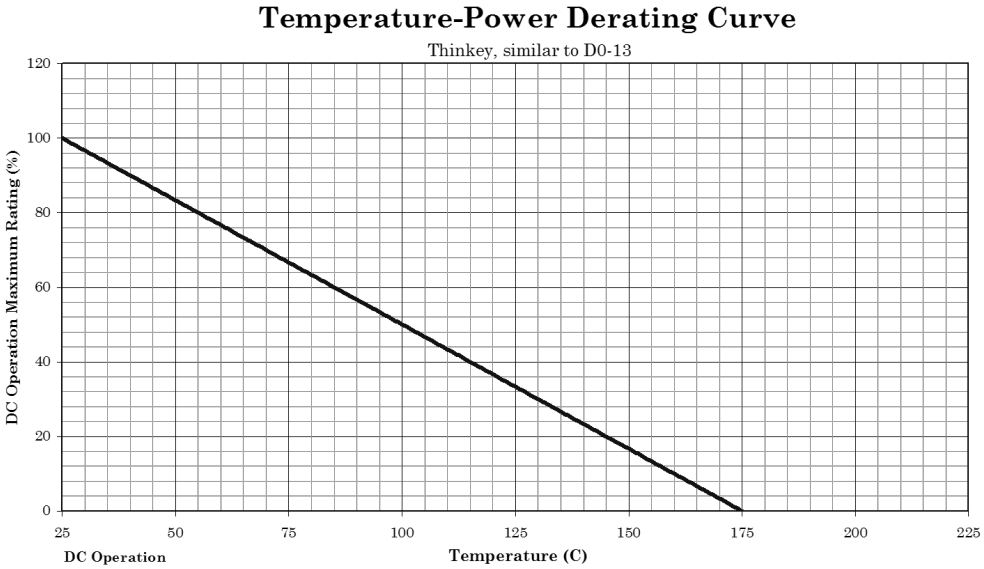



FIGURE 2. Derating curve.

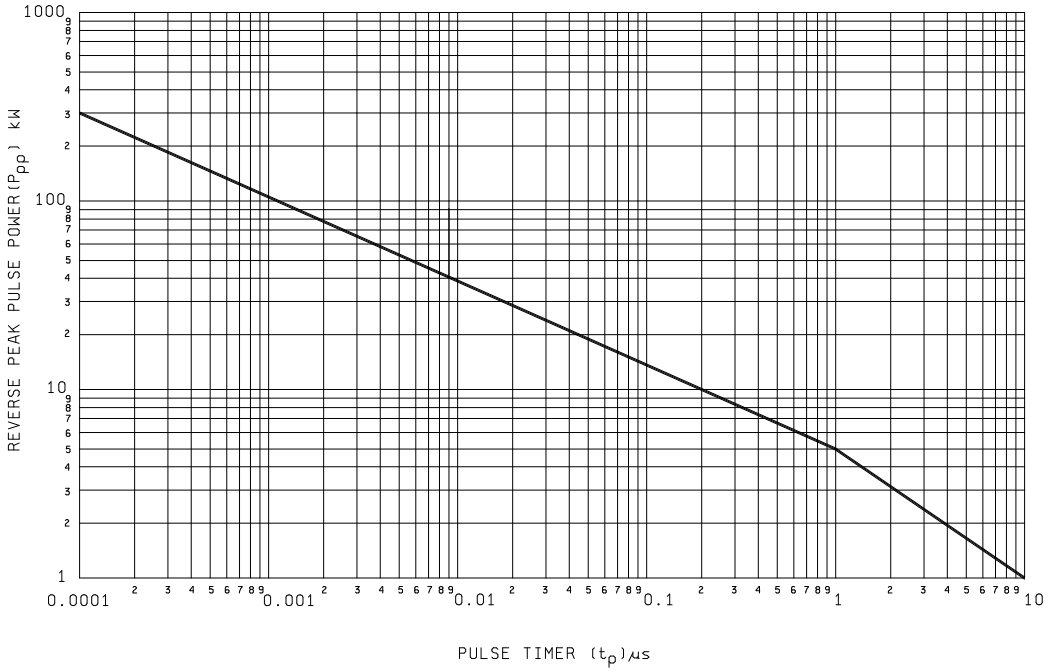
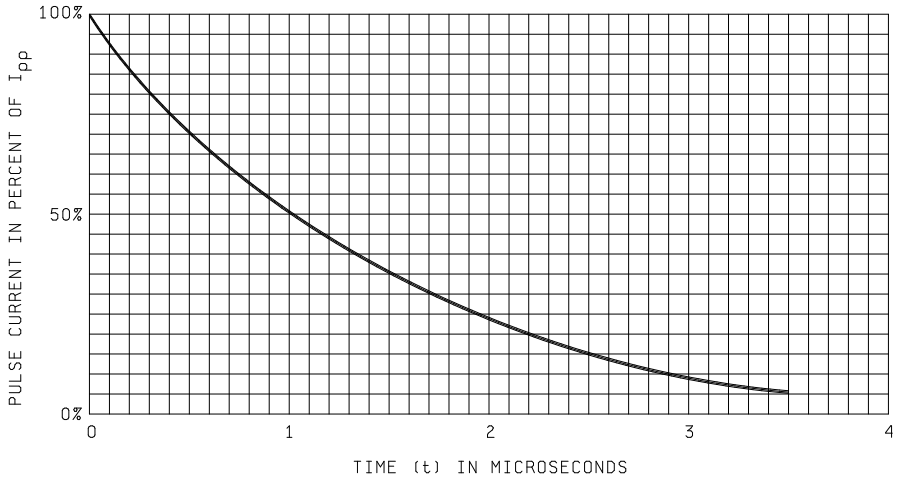



FIGURE 3. Peak pulse power versus pulse time.

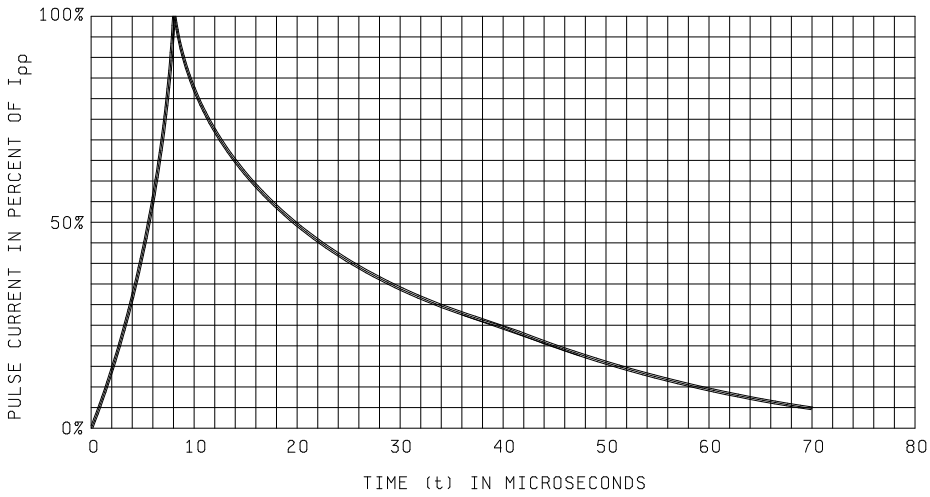


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NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{PP}$. (Rise time to 100 percent of $I_{PP}$ = 10 μs).

FIGURE 4. Current impulse waveform.




NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{PP}$. (Rise time to 100 percent of $I_{PP}$ = 8 μs).

FIGURE 5. Current impulse waveform.



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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

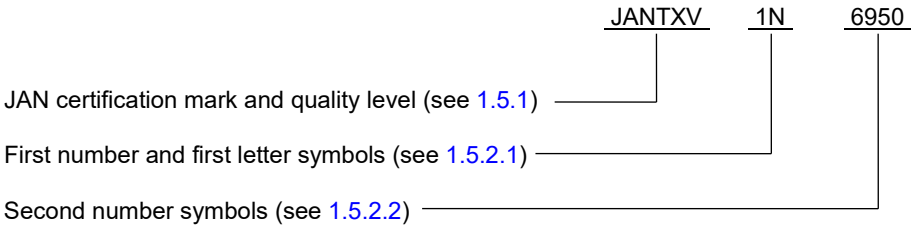
* d. Product assurance level and type designator. The complete Part or Identifying Number (PIN), see title and section 1.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: /VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 Steady state power rating. This rating is not relevant for most applications.

* 6.5 PIN construction example.

* 6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.






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* 6.6 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for types 1N6950 THROUGH 1N6986. | | | |
|---|---|---|---|
| JAN1N6950 THROUGH JAN1N6986 | JANTX1N6950 THROUGH JANTX1N6986 | JANTXV1N6950 THROUGH JANTXV1N6986 | JANS1N6950 THROUGH JANS1N6986 |

* 6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - EC
* Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - MI
* Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2019-025)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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